MANAV RACHNA
||vidyantarikshah||

L&T EduTech
Engineering the Future

MANAV RACHNA UNIVERSITY

(Department of Mechanical Engineering)

PROJECT REPORT

ON

WATERFALL MODEL FOR PRODUCT DEVELOPMENT

(SMART WATER BOTTLE)

SUBMITTED FOR

Critical Thinking, Design Thinking, Leadership and Teamwork

TASK NUMBER: 13

TOPIC: *Waterfall Models and Product Development*

SUBMITTED BY:

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Semester	:	2

SUBMITTED TO:

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INSTITUTION:

Manav Rachna University
Faridabad, Haryana

ACADEMIC YEAR:

2025 – 2026

Engineering the Future

Introduction

The Waterfall Model is a linear and sequential approach to product development where each phase must be completed before the next begins. It is simple, structured, and suitable for projects with clear requirements.

In this project, the Waterfall Model is applied to design and develop a **Smart Water Bottle** that helps users maintain proper hydration through reminders and tracking features.

Lucid Chart Link

https://lucid.app/lucidchart/a824e57c-dcf2-438b-be5e-d7e069b637ff/edit?viewport_loc=161%2C-1%2C2600%2C1063%2C0_0&invitationId=inv_4c53cab5-46a1-4b38-b832-1d429a8a65a5

2. Objective

The main objective of this project is to:

- Develop a structured product using the Waterfall Model
- Design a Smart Water Bottle that improves daily water intake
- Understand each stage of product development from planning to maintenance

3. Problem Statement

Many people forget to drink enough water during the day, leading to dehydration and health issues. There is a need for a smart solution that reminds and tracks water intake effectively.

4. Methodology – Waterfall Model Phases

4.1 Requirements Analysis

- Users forget to drink water regularly
- Need for hydration reminders
- Demand for an easy-to-use smart device
- Integration with mobile applications

Output: Defined user needs and product goals

4.2 System Design

- Bottle equipped with sensors
- LED reminder system
- Mobile app interface
- Features planned:
    - Water intake tracking
    - Notifications
    - Battery system

Output: Blueprint of system architecture

4.3 Implementation

- Development of hardware (bottle + sensors)
- Development of mobile application
- Integration of software with hardware

Output: Functional prototype

4.4 Testing

- Checking reminder accuracy
- Testing app connectivity
- Debugging errors

Output: Verified and tested system

4.5 Deployment

- Product launched into the market
- Available both online and offline
- Marketing and promotion strategies implemented

Output: Product available to users

4.6 Maintenance

- Fix bugs and issues
- Update app features
- Improve performance based on user feedback

Output: Continuous improvement of product

5. Results and Discussion

The Waterfall Model helped in clearly organizing each stage of development. The Smart Water Bottle prototype successfully addressed the problem of irregular hydration by providing reminders and tracking features.

However, the model lacks flexibility, as changes are difficult to implement once a stage is completed.

6. Advantages of Waterfall Model

- Simple and easy to understand
- Well-structured phases
- Clear documentation
- Easy to manage

7. Limitations

- Not flexible for changes
- Late testing phase
- Not suitable for complex or evolving projects

8. Conclusion

The Waterfall Model is effective for projects with well-defined requirements. In this project, it provided a clear roadmap for developing the Smart Water Bottle. While it ensures discipline and structure, modern projects may require more flexible approaches.

9. Future Scope

- Integration with IoT devices
- AI-based hydration suggestions
- Health analytics dashboard
- Smart notifications based on activity level